PART II: PATTERNS

Topic 16: Autocorrelation — The "Memory" of Data

The Core Concept

Autocorrelation means that a data point is statistically related to its own past values. In plain terms, the data has **"memory"**: if incidents are high this month, they tend to be high next month too. The conditions that produced today's number have not disappeared overnight.

Why It Matters

High autocorrelation means the system has "momentum." Understanding this prevents two common mistakes:

- **Over-reacting:** Seeing a high month and assuming a new crisis when the data is simply carrying its prior level.
- **Over-optimism:** Seeing a low month and declaring victory when the system's momentum is still upward.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	Precinct North (UoF)	Precinct South (UoF)	North Arrests
Jan	12	25	400
Feb	10	22	380
Mar	14	28	450
Apr	15	24	460
May	18	26	550
Jun	22	24	600
Jul	25	27	700
Aug	**(Missing)**	23	680
Sep	19	22	500
Oct	16	**85 ★ Outlier**	420
Nov	14	24	390
Dec	11	26	350

Precinct South (Jan–May 2023): 25, 22, 28, 24, 26. The data stays "stuck" in a narrow band. February is influenced by the conditions that existed in January; March is influenced by February. The system is self-perpetuating.

Precinct North (Jan–Jul 2023): 12, 10, 14, 15, 18, 22, 25. This series also shows high autocorrelation — each month is close to the previous — but with an upward drift (trend).

Practical implication: if autocorrelation is high, the single best predictor of next month's number is this month's number. Large changes require significant interventions — the data resists sudden shifts.